Xingyu Chen

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EDUCATION

Fudan University Shanghai, China
B.S. in Chemistry; Micro-Program in Computation and Intelligence Science Sep 2022 – Jun 2026
GPA: 3.57/4.0 | Average Score: 91/100 | Class Rank: 20/109
CS & ML Coursework: Neural Networks and Deep Learning; Natural Language Processing; Machine Learning Systems; Data Structures and Algorithms; Theory of Computation; Data Analysis; Complex Systems Theory and Application
Math Coursework: Advanced Mathematics A I–II (incl. Calculus, Linear Algebra); Multivariate Analysis; Mathematical Statistics; Applied Functional Analysis; Introduction to Stochastic Processes
Additional Coursework: Discrete Mathematics, UC San Diego Extended Studies (in progress)
Honors: Second Prize (2023–2024) and Third Prize (2022–2023), Scholarship for Outstanding Students at Fudan University; Basic Discipline Scholarship (two awards); National Silver Medal, Chemistry Olympiad
Test Scores: TOEFL iBT 5.5/6 (110/120); GRE 330 (V160, Q170)

RESEARCH EXPERIENCE

Low-Bit Quantization of Diffusion-Based VLA Policies | Fudan University Shanghai, China
Researcher | Supervisor: Prof. Weizhong Zhang Jul 2026 – Present
- Investigate W4A4/W3A3 quantization of CogACT through 768 activation interventions and seven calibration variants across four SIMPLER tasks.
- Develop and evaluate low-rank residual calibration with attention-conditioned gating and temporal alignment; use scaling controls and closed-loop evaluations to test whether offline error reductions improve policy success.

Diffusion Language Models and Flow Matching | Fudan University Shanghai, China
Researcher | Supervisor: Prof. Baojian Zhou Jul 2025 – Jan 2026
- Investigated how diffusion language models perform on planning problems.
- Presented a technical report on the mathematical formulation and physical interpretation of flow matching.

Noisy Embedding Instruction Fine-Tuning for Compact LLMs | Fudan University Shanghai, China
Co-First Author, NLP Course Research Project | Instructor: Prof. Baojian Zhou Mar 2025 – Jun 2025
- Reproduced NEFTune on TinyLlama-Chat to evaluate embedding noise for instruction fine-tuning.
- Compared noise levels on 805 AlpacaEval instructions; moderate fixed noise achieved a 64.2% pairwise win rate against standard fine-tuning, judged by DeepSeek-V3.
- Co-designed adaptive noise scheduling and Sparse-NEFT to explore regularization and embedding sparsity; embedding pruning maintained response quality close to the standard fine-tuning baseline.

Machine Learning for Chemical Reactions | Fudan University Shanghai, China
Researcher | Supervisor: Prof. Junliang Zhang Jun 2023 – Jun 2024
- Designed and trained an MLP for reaction selectivity and product-ratio prediction on in-house data ($R^2 = 0.87$).

WORK EXPERIENCE

RetailEye Shanghai, China
CTO Management Trainee – Algorithms Jun 2026–Present; Intern: May–Nov 2025
- Fine-tune Pi3 for multi-view 3D shelf reconstruction and cross-image SKU matching, achieving over 95% precision on 2,000 cross-image SKU pairs.
- Refine depth and camera poses using multi-view correspondences, robust reprojection losses, and geometric consistency; use held-out validation to assess alignment and reject unreliable updates.
- Design a retail display optimization system using linear programming to maximize freezer-layout profitability under category-concentration constraints; develop visualization modules.

Alibaba Hangzhou, China
Intern, Multimodal Scientific LLM Training Jan 2026 – May 2026
- Built a Nanotron-based distributed LLM training stack for biochemical data, supporting multiple molecular representations, multi-source ingestion, and data-lake cleaning.
- Designed a metric-driven staged pretraining and SFT pipeline with end-to-end dataloader automation and checkpointing; developed a MoT multimodal model integrating a VAE with Qwen2.5 for cross-modal fusion.

UBS Shanghai, China
Intern, Quantitative Model Development Jan 2025 – May 2025
- Used a random forest to predict bank risk-weighted assets (RWA), with SHAP values to interpret feature contributions.
- Wrote A/B tests and organized reusable functions under mentor guidance; studied derivatives pricing and XVA models.